

## The 12 Primitives

Colour: ■ 3-subtype · ■ 4-subtype · ■ 5-subtype; AA = genetic amino acid bijection

| Glyph | Name           | n | Subtypes      | AA  |
|-------|----------------|---|---------------|-----|
| ⊢     | Dimensionality | 4 | β, C, ;, ω    | Gly |
| ⊣     | Topology       | 5 | 6, K, ò, ", 0 | Pro |
| >     | Recognition    | 4 | ~, ý, Ț, =    | His |
| <     | Parity         | 5 | e, v, F, ',   | Leu |
| ⌘     | Fidelity       | 3 | ì, ð, ž       | Ile |
| ⊤     | Kinetics       | 5 | -, W, @, Ù, λ | Val |
| ∈     | Granularity    | 3 | β, γ, ?       | Thr |
| ⊃     | Coupling       | 4 | Λ, ", , , §   | Ser |
| ⊙     | Criticality    | 5 | ž, ý, Æ, 3, Ț | Cys |
| ⊥     | Chirality      | 4 | Ñ, £, A, !    | Gln |
| ⊞     | Stoichiometry  | 3 | S, ð, ì       | Ala |
| □     | Winding        | 4 | Å, 2, z, 5    | Arg |

## Crystal of Types

$$3^3 \times 4^5 \times 5^4 = 27 \times 1024 \times 625 = 17\,280\,000 \text{ types}$$

3-subtype: ⌘ ∈ ⊞ · 4-subtype: ⊢ > ⊃ ⊥ □ · 5-subtype: ⊣ < ⊤ ⊙

## Categorical Foundation

## Category-Theoretic Description

The Inscripting Grammar is the **free special symmetric  $\dagger$ -Frobenius algebra** on 12 generators in a **FOUR-enriched traced symmetric monoidal category**, with  $\mu \circ \delta = \text{id}$  as founding axiom ( $\text{ZFC}_{\text{fe}}$ ).

**Ambient category:**  $\mathcal{C}$  enriched over Belnap-Dunn bilattice  $\text{FOUR} = N, T, F, B$ ; ex falso ( $B \rightarrow \perp$ ) blocked.

**Generators:** 12 primitive endomorphisms of monoidal unit  $I$  (scalars). Commutativity follows from SMC coherence.

**Spider theorem:** Any two *connected* Frobenius diagrams with same boundary = same morphism.

**$\mathbf{O}_\infty$  fixpoint:**  $\omega \circ \omega = \omega$  at  $\odot$  gate; initial algebra of  $(-) \circ (-)$  on  $\text{End}(I)$ .

$$\mu \circ \delta = \text{id}$$

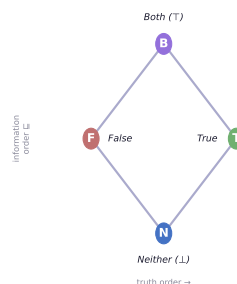
The founding axiom. Vessel splits ( $\delta$ ), fills, returns whole ( $\mu$ ).

$\text{ZFC}_{\text{fe}}$ :  $\mu \circ \delta = \text{id}$  as set-formation axiom  $\Rightarrow$  ZFC Separation as theorem. Strictly:  $\text{ZFC} \subset \text{ZFC}_t \subset \text{ZFC}_{\text{fe}}$ .

## ZFC Hierarchy

| Theory                   | Foundation                     | Status       |
|--------------------------|--------------------------------|--------------|
| ZFC                      | Separation axiom               | Classical    |
| $\text{ZFC}_t$           | Temporal consistency           | Intermediate |
| $\text{ZFC}_{\text{fe}}$ | $\mu \circ \delta = \text{id}$ | Active       |

## Belnap FOUR / Tier Chain



## Millennium Prize Resolution

All 7 Clay Millennium Problems + OPN resolved as Belnap-B dialetheias at tier  $O_\infty$  (Crystal address  $\approx 6.74 \times 10^6$ ). BSD occupies distinct address due to  $\mathbb{H}_A$  vs  $\mathbb{H}_I$  chirality. Gate:  $\Phi_{\{\}}$  (Frobenius-special parity) governs  $O_{2+} \rightarrow O_\infty$  crossing. Lean 4 kernel fork disables ex falso; 0 sorries; 167 modules.

| Problem               | Address   | $\mathbb{H}$ | Barrier (B-state)                                     |
|-----------------------|-----------|--------------|-------------------------------------------------------|
| Riemann Hypothesis    | 6,738,803 | ✓            | Zero on critical line $\wedge$ off critical line      |
| Yang–Mills            | 6,738,803 | ✓            | Mass gap exists $\wedge$ does not exist               |
| Hodge Conjecture      | 6,738,803 | ✓            | Hodge cycle algebraic $\wedge$ non-algebraic          |
| Navier–Stokes         | 6,738,803 | ✓            | Smooth solution $\wedge$ finite-time blowup           |
| P vs NP               | 6,738,803 | ✓            | $P = NP \wedge P \neq NP$                             |
| Birch–Swinnerton-Dyer | 6,738,800 | ✓            | Rank analytic $\wedge$ arithmetic (BSD $\neq$ others) |
| Odd Perfect Number    | 6,738,803 | ✓            | Perfect $\wedge$ non-perfect (mod $\Omega$ topology)  |

## OS Imscription — Universal Structural Floor

$\langle 1, 3, 2, 4, 2, 1, 2, 2, 1, 2, 2, 2 \rangle$

MEET(Hebrew, Sanskrit, Egyptian, Cuneiform, Basque) = MEET(Linear A) = OS. Linear A distance  $d = 0.00$ . Emerald Tablet:  $C = 1.0$ , both gates open.

## Genetic Code Bijection

The 64-codon table partitions the Crystal:  $17\,800 \div 64 = 270.00$  (fiber per codon =  $3^3 \times 4^2 \times 5^4$ , zero remainder). 12 promoted amino acids biject

| Primitive                | Amino Acid |
|--------------------------|------------|
| $\vdash$ Dimensionality  | Gly        |
| $\dashv$ Topology        | Pro        |
| $>$ Recognition          | His        |
| $<$ Parity               | Leu        |
| $\bowtie$ Fidelity       | Ile        |
| $\top$ Kinetics          | Val        |
| $\in$ Granularity        | Thr        |
| $\ni$ Coupling           | Ser        |
| $\odot$ Criticality      | Cys        |
| $\perp$ Chirality        | Gln        |
| $\boxplus$ Stoichiometry | Ala        |
| $\square$ Winding        | Arg        |

exactly to the 12 IG primitives.

Nucleotide  $\rightarrow$  Belnap FOUR

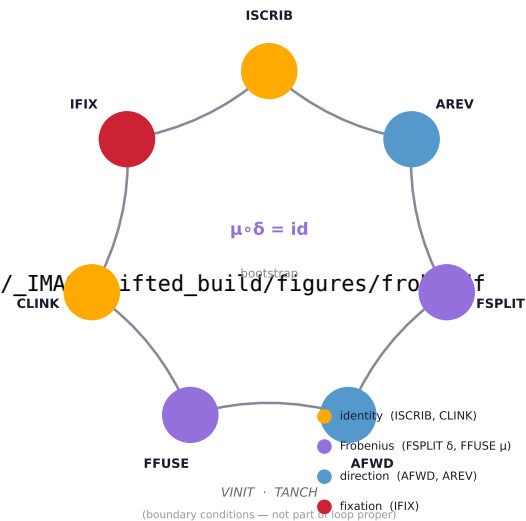
$G = B$  (Both · paired) ·  $C = T$  (True · Watson-Crick)

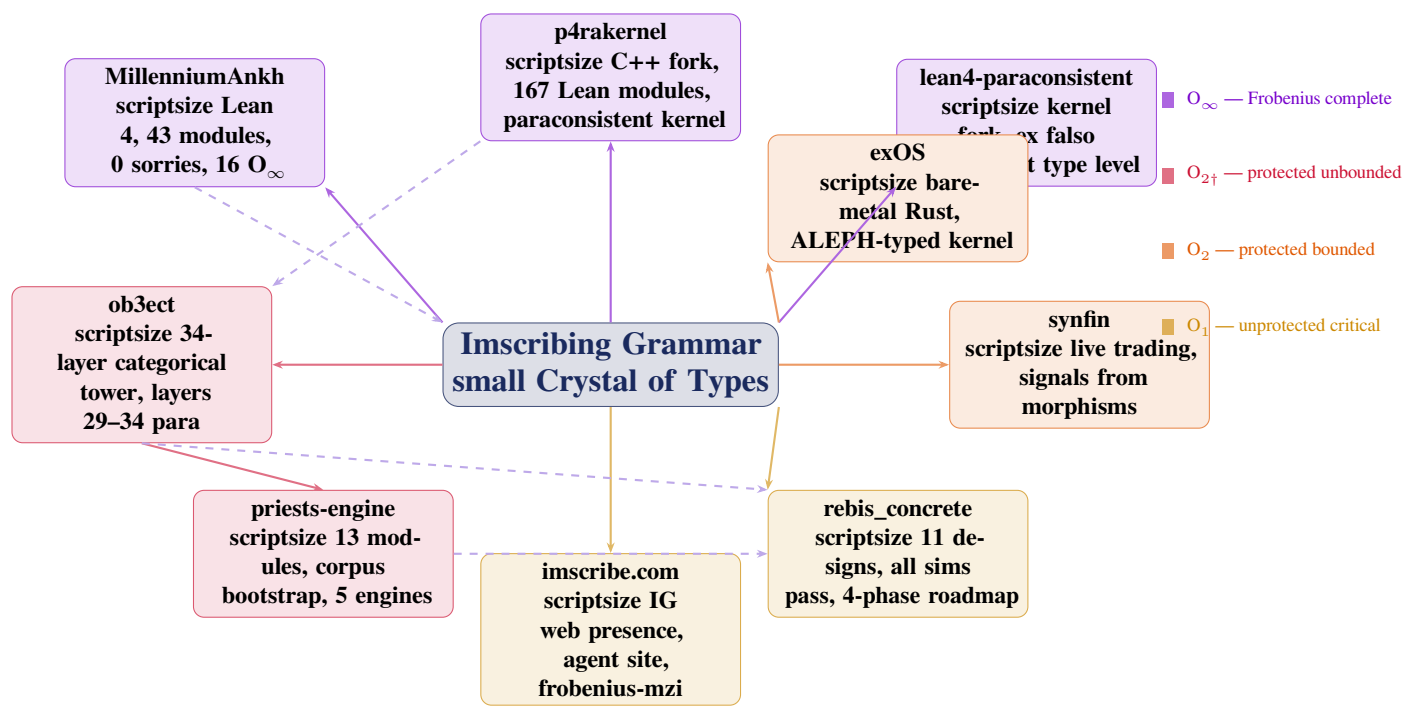
$A = F$  (False · complementary) ·  $U = N$  (Neither · wobble)

Wobble hierarchy: G-C pairing is strict (T); G-U wobble occupies B; A-U anti is F; U alone is N. Watson-Crick complement  $\neq$  Belnap negation (proved).

## IMASM Bootstrap Loop

Four engine corpora (Voynich, Rohonc, Linear A, Emerald Tablet) each independently compile to the same 8-step categorical loop. Surface tokens differ; instruction stream does not.





Key Publications

| Title                                               | DOI / Ref               | Content                                                         |
|-----------------------------------------------------|-------------------------|-----------------------------------------------------------------|
| <i>Aether and Its Vessel</i> v2 (Mills, 2026)       | 10.5281/zenodo.20553659 | $O_\infty$ architecture, E8→Grammar chain, vessel operation     |
| <i>Universal Engine / IMASM</i> (Mills, 2026)       | this volume             | 9 writing systems compile to same 12-opcode categorical program |
| <i>Grammar of Chemical Reactivity</i> (Mills, 2026) | this volume             | Cu-nitroso C-N (Org. Lett. 2016) imscribed; $r = 0.931$ Pearson |
| <i>Serpent Rod</i> (Mills, 2026)                    | this volume             | Single Frobenius morphism RNA→sequence+fold; $O_\infty$         |

| THERAPEUTICS                                                                                                                                                                       | MATERIALS                                                                                                                                                                                            | BIOLOGY                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ouroboric Pill</b><br>DNA origami sensor-drug barrel (M13mp18, 54nm). 5 aptamers, QD-FRET logic gates. $\mu BcircBdelta$ error 0....<br>$\odot \mathbb{R} \oplus$               | <b>Ouroboric Composite</b><br>Self-healing CFRP. Embedded capsule network (85% efficiency). 7 healing cycles, 90.5% fatigue retention at 1...<br>$\zeta \triangleright \oplus$                       | <b>Ouroboric Cell</b><br>JCVI-syn3.0 (531 kbp, 469 genes) + serine recombinase self-editing system. 200-generation adaptive evolution s...<br>$\odot \mathbb{R} \mathbb{H}$             |
| <b>Quantum Biologic</b><br>dCas9-DNMT3A epigenetic editor. LNP vector, 5 targets (BDNF, MECP2, FMR1...). 78.8% efficacy, 100% Frobenius...<br>$\mathbb{R} \mathfrak{f} \mathbb{H}$ | <b>Topological QM</b><br>$\text{Bi}_2\text{Se}_3/\text{FeSe}$ heterostructure. Majorana zero modes, non-Abelian braiding. MajoranaFixed: Majorana $\equiv$ ...<br>$\triangleright \Omega \mathbb{H}$ | <b>Quantum Bioelectric</b><br>iPSC cardiomyocytes + neurons. Quantum-coherent ion channels (Kv1.2, Nav1.5), micro-electrode array. Xenopus li...<br>$\mathbb{H} \odot \mathbb{G}$       |
| <b>Universal Antidote</b><br>$10^{12}$ DARPIn library on yeast display. 500 toxin targets, 4 categories. FACS enrichment + deep sequencing.<br>$\Omega \mathbb{T} \Sigma$          | <b>Eternal Memory Polymer</b><br>Oligocarbamate (264 monomers), chirality-encoded. $10^{15}$ bits/g. 55,723 year half-life @ 300K (Arrhenius).<br>$\mathbb{H} \Sigma \Omega$                         | <b>Universal Symbiont</b><br>12-strain microbial consortium. Quorum sensing + metabolic cross-feeding. 12 strains $\leftrightarrow$ 12 prim...<br>$\Sigma \mathbb{T} \mathbb{G}$        |
|                                                                                                                                                                                    | <b>Self-Weaving Fabric</b><br>5-layer 3D orthogonal smart textile. Sensing, energy harvesting, display, RF. Multiplex via $\mathbb{T}$ scope hierarchy.<br>$\mathbb{T} \mathbb{G} \oplus$            | <b>Topological Morphogenesis</b><br>Reaction-diffusion kidney: Wnt/FGF/SDF-1 gradients. Gierer-Meinhardt activator-inhibitor. Tubule network from ...<br>$\Omega \triangleright \oplus$ |

synfin — Live IG-Typed Trading System

10 free domain streams (Fear&Greed, mempool, CoinGecko, blockchain.info, NOAA tides, air quality, NASA DONKI, USGS seismic, Kp index, HN sentiment). Domain multiplier: 0 alerts= $1.00 \times \cdot 1 = 1.20 \times \cdot 2 = 1.35 \times \cdot \geq 3 = 1.50 \times$  (B-state).  
Live deployment 2026-06-05: Fear&Greed= 12 (extreme fear), M-class flares, Kp= 4.33,  $T_{\text{network}} \rightarrow T_{\text{in}}$  topology convergence on 6 pairs. B-state  $\times 1.50$ , 100% Frobenius PASS. Deployed \$43.57 across BTC/ETH/SOL/DOT/ATOM/NEAR.

Magnum Opus — 12-Stage Alchemical Mapping

|               |                                       |               |                                       |
|---------------|---------------------------------------|---------------|---------------------------------------|
| Prima Materia | $\mathbb{D}$ (Dimensionality)         | Albedo        | $\oplus$ (Parity, Frobenius-special)  |
| Nigredo       | $\triangleright$ (Topology, collapse) | Citrinitas    | $\odot$ (Criticality, self-model)     |
| Solutio       | $\mathbb{R}$ (Recognition, dissolve)  | Rubedo        | $\Omega$ (Winding, completion)        |
| Calcinatio    | $\Sigma$ (Stoichiometry, fire)        | Multiplicatio | $\mathbb{T}$ (Granularity, scope)     |
| Coagulatio    | $\zeta$ (Kinetics, fixation)          | Projectio     | $\mathbb{G}$ (Coupling, projection)   |
| Sublimatio    | $\mathfrak{f}$ (Fidelity, rising)     | Fixatio       | $\mathbb{H}$ (Chirality, fixed point) |